Note that although $\lim_{x\to -4} f(x)$ exists, the function value at -4, namely $f(-4)$ is not defined. Thus the existence of $\lim_{x\to -4} f(x)$ has no bearing on the existence of $f(-4)$.

Now we formally define continuity as in

Definition 9.7

Let I be an open interval in $\mathbb{R}$ containing x_0. Let $f : I \to \mathbb{R}$. Then f is said to be continuous at x_0 if it is defined in a neighbourhood of this point and if the limit of this function, as the independent variable x tends to x_0, exists and is equal to the value of the function at $x = x_0$.

Thus three requirements have to be satisfied for the continuity of a function $y = f(x)$ at $x = x_0$:

(i) $f(x)$ must be defined in a neighbourhood of x_0 (i.e., $f(x_0)$ exists);

(ii) $\lim_{x\to x_0} f(x)$ exists ;

(iii) $f(x_0) = \lim_{x\to x_0} f(x)$.

The condition (iii) can be reformulated as $\lim_{\Delta x\to 0}\left[f(x_0+\Delta x) - f(x_0)\right] = 0$ and the continuity of f at x_0 can be restated as follows :

Definition 9.8

A function $y = f(x)$ is said to be continuous at a point x_0 (or at $x = x_0$) if it is defined in some neighbourhood of x_0 and if $\lim_{\Delta x\to 0}\left[f(x_0+\Delta x) - f(x_0)\right] = 0$.

The condition (iii) can also be put in the form $\lim_{x\to x_0} f(x) = f\left(\lim_{x\to x_0} x\right)$. Thus, if the symbol of the limit and the symbol of the function can be interchanged, the function is continuous at the limiting value of the argument.

9.3.1 Examples of functions Continuous at a point

(1) Constant function is continuous at each point of $\mathbb{R}$.

Let $f(x) = k, k \in \mathbb{R}$ is constant. If $x_0 \in \mathbb{R}$, then $f(x_0) = k$.

$$\lim_{x\to x_0} f(x) = \lim_{x\to x_0}(k) = k.$$

(2) Power functions with positive integer exponents are continuous at every point of $\mathbb{R}$

If $f(x) = x^n$, domain of f is $\mathbb{R} = (-\infty, \infty)$ and $\lim_{x\to x_0} x^n = x_o^n, x_0 \in \mathbb{R}$ by the limit theorem.

(3) Polynomial functions, $p(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n, \quad a_0 \neq 0$

are continuous at every point of $\mathbb{R}$. By limit theorem,

$$\lim_{x\to x_0} p(x) = a_0x_0^n + a_1x_0^{n-1} + \cdots + a_{n-1}x_0 + a_n = p(x_0).$$

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